

OS Project 1 - Project Report

Counting Primes

A. Design and Implementation

- a. There are two structs, Job and Result. Result extends Job with an additional field for the prime count. Channels store these struct variables.
 - b. Sequence of initializing goroutines: i. Single dispatcher ii. M workers iii. Single consolidator
 - c. The dispatcher partitions the input file into segments of N bytes and pushes job descriptors into a buffered jobs channel. It uses `defer close(jobsCh)` to guarantee the channel is always closed even on error.
 - d. Workers read job descriptors from the jobs channel. Each worker sleeps for a random 400-600ms on startup, then reads its assigned file segment in chunks of C bytes, decodes 64-bit unsigned integers using the encoding/binary package with little-endian byte order, and counts prime numbers using Go's built-in probabilistic Miller-Rabin primality test via `math/big.ProbablyPrime(20)`. Results are pushed into a buffered results channel.
 - e. The consolidator reads partial results from the results channel, accumulates the prime counts, and sends the final total to `main()`. A `sync.WaitGroup` tracks active workers — when all workers finish, `resultsCh` is closed, signaling the consolidator to stop.
 - f. Minimum information is communicated between threads — only job descriptors (pathname, start, length) and result descriptors (job + prime count).
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B. Q1 — Largest Number of Workers M Supported

My implementation supports a maximum of approximately **10,000,000** workers on my personal computer. M=10,000,000 completed successfully while M=20,000,000 caused the system to hang due to memory exhaustion.

- Command: `go run main.go test_10MB.dat 10000000 65536 8192`
- M: 10,000,000 | N: 65536 | C: 8192

- Min jobs completed by a worker: 0
- Max jobs completed by a worker: 1
- Average jobs completed by a worker: 0.0
- Median jobs completed by a worker: 0.0
- Elapsed time (ms): 10,153 ms

Sample output:

None

```
pathname: test_10MB.dat, M: 10000000, N: 65536, C: 8192
Total prime numbers found: 30546
Min jobs completed by a worker: 0
Max jobs completed by a worker: 1
Average jobs completed by a worker: 0.0
Median jobs completed by a worker: 0.0
Elapsed time (ms): 10153 ms
```

C. Q2 — Least Elapsed Time for a 1GB Datafile

The least elapsed time for a random 1GB datafile in my implementation was **66,086 ms** with M=16.

- Command: go run main.go test_1GB.dat 16 65536 8192
- M: 16 | N: 65536 | C: 8192
- Min jobs completed by a worker: 1016
- Max jobs completed by a worker: 1033
- Average jobs completed by a worker: 1024.0
- Median jobs completed by a worker: 1024.0
- Elapsed time (ms): 66,086 ms

Sample output:

None

```
pathname: test_1GB.dat, M: 16, N: 65536, C: 8192
Total prime numbers found: 3096150
Min jobs completed by a worker: 1016
Max jobs completed by a worker: 1033
Average jobs completed by a worker: 1024.0
```

```
Median jobs completed by a worker: 1024.0
Elapsed time (ms): 66086 ms
```

D. Q3 — Largest Datafile Processed Within 3 Minutes

The largest datafile processed within 3 minutes elapsed time was **2GB**. The 2.5GB file was tested and completed in 198,482ms (~3.3 minutes), slightly exceeding the limit. The 2GB file completed in 134,614ms (~2.24 minutes).

- Command: go run main.go test_2048MB.dat 16 65536 8192
- M: 16 | N: 65536 | C: 8192
- Min jobs completed by a worker: 2037
- Max jobs completed by a worker: 2057
- Average jobs completed by a worker: 2048.0
- Median jobs completed by a worker: 2048.0
- Elapsed time (ms): 134,614 ms

Sample output:

```
None
pathname: test_2048MB.dat, M: 16, N: 65536, C: 8192
Total prime numbers found: 6192296
Min jobs completed by a worker: 2037
Max jobs completed by a worker: 2057
Average jobs completed by a worker: 2048.0
Median jobs completed by a worker: 2048.0
Elapsed time (ms): 134614 ms
```

E. Q4 — Elapsed Time vs Number of Workers M

Using test_1GB.dat with N=65536, C=8192:

M	Elapsed Time (ms)	Min Jobs	Max Jobs	Avg Jobs	Median Jobs
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1	387,126	16,384	16,384	16,384	16,384
2	230,432	8,190	8,194	8,192	8,192
4	137,162	4091	4099	4096.0	4097.0
8	81,471	2044	2052	2048.0	2048.5
12	71,810	1360	1369	1365.3	1365.5
16	66,086	1016	1033	1024.0	1024.0
32	66,949	499	523	512.0	511.5

Sample outputs:

M=1:

None

```
pathname: test_1GB.dat, M: 1, N: 65536, C: 8192
Total prime numbers found: 3096150
Min jobs completed by a worker: 16384
Max jobs completed by a worker: 16384
Average jobs completed by a worker: 16384.0
Median jobs completed by a worker: 16384.0
Elapsed time (ms): 387,126 ms
```

M=4:

None

```
pathname: test_1GB.dat, M: 2, N: 65536, C: 8192
Total prime numbers found: 3096150
Min jobs completed by a worker: 8190
Max jobs completed by a worker: 8194
Average jobs completed by a worker: 8192.0
Median jobs completed by a worker: 8192.0
Elapsed time (ms): 230432 ms
```

M=4:

None

```
pathname: test_1GB.dat, M: 4, N: 65536, C: 8192
Total prime numbers found: 3096150
Min jobs completed by a worker: 4091
Max jobs completed by a worker: 4099
Average jobs completed by a worker: 4096.0
Median jobs completed by a worker: 4097.0
Elapsed time (ms): 137162 ms
```

M=8:

None

```
pathname: test_1GB.dat, M: 8, N: 65536, C: 8192
Total prime numbers found: 3096150
Min jobs completed by a worker: 2044
Max jobs completed by a worker: 2052
Average jobs completed by a worker: 2048.0
Median jobs completed by a worker: 2048.5
Elapsed time (ms): 81471 ms
```

M=12:

None

```
pathname: test_1GB.dat, M: 12, N: 65536, C: 8192
Total prime numbers found: 3096150
Min jobs completed by a worker: 1360
Max jobs completed by a worker: 1369
Average jobs completed by a worker: 1365.3
Median jobs completed by a worker: 1365.5
Elapsed time (ms): 71810 ms
```

M=16:

None

```
pathname: test_1GB.dat, M: 16, N: 65536, C: 8192  
Total prime numbers found: 3096150  
Min jobs completed by a worker: 1016  
Max jobs completed by a worker: 1033  
Average jobs completed by a worker: 1024.0  
Median jobs completed by a worker: 1024.0  
Elapsed time (ms): 66086 ms
```

M=32:

None

```
pathname: test_1GB.dat, M: 32, N: 65536, C: 8192  
Total prime numbers found: 3096150  
Min jobs completed by a worker: 499  
Max jobs completed by a worker: 523  
Average jobs completed by a worker: 512.0  
Median jobs completed by a worker: 511.5  
Elapsed time (ms): 66949 ms
```

As M increases from 1 to 16, elapsed time decreases significantly. Beyond 16 workers the gains diminish as M exceeds the number of physical CPU cores (12 cores on this machine), and scheduling overhead begins to offset the parallelism benefits.